q6_loan.c - Code::Blocks / VS Code (C)

```c
#include <stdio.h>

int main(void)
{
    double balance = 10000.0, rate = 0.02, pay = 2000.0, total_interest = 0.0;
    const int LIMIT = 360;
    int month = 0;

    if (pay <= balance * rate) {
        printf("Repayment %.2f <= monthly interest %.2f: loan can never be repaid.\n",
               pay, balance * rate);
        return 1;
    }

    printf("Month  Interest   Payment   Balance\n");
    while (balance > 0.005) {
        if (month >= LIMIT) {
            printf("Stopped: not repaid after %d months. Review the loan terms.\n", LIMIT);
            break;
        }
        double interest = balance * rate;
        balance += interest;
        total_interest += interest;
        double payment = (balance < pay) ? balance : pay;
        balance -= payment;
        month++;
        printf("%5d %9.2f %9.2f %9.2f\n", month, interest, payment, balance);
    }
    printf("Months = %d, total interest = %.2f\n", month, total_interest);
    return 0;
}
```

Terminal - output

```
user@pc:~/Assignment2$ gcc -Wall q6_loan.c -o q6 && ./q6
Month  Interest   Payment   Balance
    1    200.00   2000.00   8200.00
    2    164.00   2000.00   6364.00
    3    127.28   2000.00   4491.28
    4     89.83   2000.00   2581.11
    5     51.62   2000.00    632.73
    6     12.65    645.38      0.00
Months = 6, total interest = 645.38
user@pc:~/Assignment2$ _
```